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Research Article

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Integration of mathematical modeling and target-based application of biocontrol agents for the control of *Botrytis cinerea* in vineyards

Valeria Altieri, Vittorio Rossi and Giorgia Fedele*

Abstract

BACKGROUND: Biocontrol agents (BCAs) are alternatives to synthetic fungicides with low risk to the environment and human health. Although several studies on the biocontrol of gray mold in vineyards have been performed, it is necessary to improve the usage of BCAs in fields conditions. Therefore, in the present study, BCAs were used both in calendar-based [based on four growth stages (GSs), i.e., flowering, pre-bunch closure, veraison, and before harvest] and predictive model-based strategies (only when *Botrytis cinerea* infection risk was predicted by the model). The BCAs applied during the seasons were selected considering the grapevine GSs. Treatments performed with BCAs were compared with synthetic fungicide treatments and an untreated control. The trials were conducted in three experimental vineyards with four epidemics. To evaluate the level of gray mold control of each treatment, disease severity was assessed at harvest and the presence of latent infection was evaluated.

RESULTS: The integrative use of the predictive model and BCAs provided satisfactory levels of gray mold control, with gray mold severity levels significantly lower ($P < 0.001$) than those of the untreated control, which had severity values (< 7%) similar to those observed with synthetic fungicides following both calendar and model-based strategies.

CONCLUSIONS: The integrative use of the predictive model and BCAs represents a valid alternative to conventional methods of gray mold control in vineyards, with more than 75% reduction in fungicide usage.
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Keywords: biological control; biological control agents; microorganisms; gray mold; mechanistic weather-driven model

1 INTRODUCTION

Grapevine gray mold, caused by *Botrytis cinerea*, can cause serious damage to the yield and quality of grapes.[1] Given the seriousness of the disease, the traditional *Botrytis cinerea* control in vineyards is commonly based on a routine calendar application of chemical fungicides at four specific grape growth stages (GSs; i.e., flowering, pre-bunch closure, veraison, and before harvest).[2] This application schedule often leads to unnecessary sprays as it does not take into account the real risk of *Botrytis cinerea* infection.[2] Additionally, the massive use of synthetic fungicides has resulted in the development of fungicide-resistant strains of *Botrytis cinerea* within vineyard populations[3,4] along with negative impacts on the environment and human health.[5,6]

To support the agriculture transition toward more sustainable production systems, new regulatory frameworks have been developed to promote integrated pest management and the use of advanced tools and alternatives to chemicals has been suggested.[7] Particularly, decision tools, including plant protection models, could play a key role in defining which and when plant protection products are needed for disease control.[8]

For the control of grapevine gray mold, a mechanistic, weather-driven model was developed by González-Domínguez *et al.*,[9] which was implemented in a decision support system called Vite.net®.[10] The model accounts for multiple *Botrytis cinerea* infection paths, including inflorescences, young berries, and maturing berries infection by conidia, and berry-to-berry infection from berries touching each other during harvest.[1,9] Weather conditions and host GSs can also be considered to address the infection risk in the two periods of grapevine susceptibility, that is, between flowering to young cluster development and after majority of berries touching each other.[9] The model, validated with a total of 44 epidemics,[9,11] represents an advanced tool for the daily prediction of *Botrytis cinerea* infection, which can help growers to decide whether a fungicide application is needed. Furthermore, the use of the model ensures the transition from a calendar-based to a model-based disease management approach. The latter has been widely implemented for the application of synthetic fungicides to manage common diseases in viticulture[12,13] and less extended to the use of alternative products, including biocontrol agents (BCAs).

BCAs are certain microorganisms or microorganism-based plant protection products, which can be used as alternatives to chemical fungicides for the control of gray mold.[5,14–16] Despite the progressive increase in the availability of BCAs in the market[17,18] and

* *Correspondence to: G Fedele, Department of Sustainable Crop Production, DiProVeS, Università Cattolica del Sacro Cuore di Piacenza, via Emilia Parmense 84, 29122 Piacenza, Italy. E-mail: giorgia.fedele@unicatt.it*

Department of Sustainable Crop Production, DiProVeS, Università Cattolica del Sacro Cuore di Piacenza, Piacenza, Italy

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extensive research on gray mold biocontrol, the practical use of BCAs is still limited due to some shortcomings,[19] which result in high variability of their field efficacy.[15,20–22] These shortcomings are related to (i) the complexity of the life cycle of *Botrytis cinerea*, which results in multiple infection pathways,[1] affecting different plant parts (mainly flowers and young fruit, bunch trash, and berries at different ripening stages) and different plant GSs (flowering, pre-bunch closure, and ripening), (ii) the diversity of the mode of action (MoA) of different biocontrol microorganisms, (iii) different attitude of different BCAs to grow on different plant parts,[23–25] and (iv) their specific ecological requirements.[22,26,27]

To address these shortcomings, Pertot *et al.*[15] and Fedele *et al.*[28] proposed to apply different BCAs at different plant GSs at which disease control interventions are classically applied, that is, at flowering, pre-bunch closure, veraison, and ripening.[2]

In the present study, we developed the concept of a target-based selection of BCAs for the control of *Botrytis cinerea* in vineyards and applied the selected BCAs based on the infection risk predicted by the model of González-Domínguez *et al.*[9] We compared the efficacy of model-based treatments with that of calendar-based application of either BCAs or chemical fungicides to verify whether this innovative approach can efficiently control gray mold and simultaneously save unjustified sprays in four *Botrytis cinerea* epidemics.

2 MATERIALS AND METHODS

2.1 Experimental vineyards

Four epidemics (i.e., a combination of a vineyard in a year) were considered in 2022 and 2023 in three vineyards located in the Emilia-Romagna region (Italy): (i) Castell'Arquato (CA) (44° 51′26.1″N, 9°51′20.7″E), (ii) Piacenza (PC) (45°2'N, 9°43′E), and (iii) Ravenna (RA) (44°52′68.3″N 12°18′40.9″E). The experiment in CA was performed both in 2022 and 2023 (labeled as CA22 and CA23, respectively), whereas in PC and RA, the experiment was conducted only in 2023 (labeled as PC23 and RA23, respectively). The plot position in CA vineyard varied among years to reduce possible interference between the two epidemics. The vineyard in CA was planted in 2009 with the cultivar Barbera and trained using the Guyot system; within- and between-row spacings in the vineyard were 1.3 and 2.4 m, respectively. The vineyard in RA was planted in 2000 with the cultivar Merlot and trained using the Geneva Double Curtain trellis system; within- and between-row spacings were 1.2 and 2.5 m, respectively. The vineyard in PC was planted in 2011 with the cultivar Merlot and trained using the Guyot system; within- and between-row spacings were 1.2 and 2 m, respectively. Barbera and Merlot are red cultivars with compact bunches and thin skin and are known to be highly susceptible to *Botrytis cinerea*.[14,29] All three vineyards were managed following common practices and only fungicides without efficacy against *Botrytis cinerea* were used to control downy and powdery mildews.

2.2 Model structure and running

The model has been described by González-Domínguez *et al.*[9] and included various grape GSs. In brief, the model begins to operate when grape inflorescences are clearly visible (GS53) and ends at harvest (GS90), with a time step of 1 day. The model assumes that the fungal inoculum sources are present in the vineyard and estimates the relative abundance of conidia produced on inoculum sources on any day of the grape-growing season as a function of the rate at which the mycelium grows and saprophytically colonizes the source tissue and produces conidia. The model then assumes that on any day, mature conidia may disperse and settle on host plant surfaces. The model considers two main infection periods. In the first infection period [between GS53 (inflorescences clearly visible) and GS73 (berries groat-sized, bunches begin to hang)], the model calculates a daily infection risk of inflorescences and young clusters by conidia (pathways I and II).[1] In the second infection period [between GS79 (majority of berries touching each other) and GS89 (berries are ripe for harvest)], the model calculates the daily infection risk of ripening clusters by conidia (pathway Va)[1] and berry-to-berry infection by aerial mycelium (pathway Vb).[9] Daily values of infection risk are accumulated over the corresponding infection period and accumulated values produce new variables (SEV1, SEV2, and SEV3), which contribute to the total risk of infection. The model finally uses SEV1, SEV2, and SEV3 as independent variables in a discriminant function analysis to classify the epidemics into three groups based on disease severity (DS) at harvest: severe, (DS ≥ 15%), intermediate (5 < DS < 15%), or mild (DS ≤ 5%). For each vineyard, the model was operated using the vineyard's weather data and grapevine GSs to predict, on any day of the vine-growing season, the epidemic group (mild, intermediate, or severe) at harvest.[9]

2.3 Treatments

Calendar- and model-based applications of both chemical fungicides (hereinafter named CHEM) and BCAs for the control of gray mold were compared; a non-treated (NT) control was also included. The calendar-based schedule comprised routine fungicide application at four grape GSs: (i) flowering (A; GS65[30]), (ii) pre-bunch closure (B; GS77), (iii) veraison (C; GS83), and (iv) before harvest (D; before GS89). The model-based schedule was based on suggestions provided by the predictive model described earlier,[9] in which CHEM or BCAs were applied only when a severe disease risk was predicted. Following a fungicide resistance management strategy,[31] fungicides were applied as follows: Fluazinam was applied at stage A, Cyprodinil + Fludioxonil at stage B, Boscalid at stage C, and Cyprodinil + Fludioxonil at stage D considering a preharvest interval of 21 days (Table 1). In connection with BCAs, *Aureobasidium pullulans* DMS 14941-14940 was used at stage A, *Trichoderma atroviride* SC1 (in 2022) and a combination of *Trichoderma gamsii* ICC 080 and *Trichoderma asperellum* ICC 012 (in 2023) at stage B, *Bacillus amyloliquefaciens* D747 at stage C, and *Metschnikowia fructicola* NRRL Y-27328 at stage D.

Products were applied at the manufacturer-recommended doses (Table 1) using a 15-L Elettroplus knapsack sprayer (Davide and Luigi Volpi S.p.a, Casalromano, Italy) following a complete randomized block design with four replicate plots (six plants per plot). Hourly data of temperature, relative humidity (RH), wetness duration, and rainfall were recorded by automated weather stations (iMeteos; Pessl Instruments GmbH, Weiz, Austria) located at a distance of < 1 km from the experimental plots and were used to run the model.

2.4 Disease assessments

The GSs of vines were assessed twice a week from clearly visible inflorescences (GS53) until harvest (> GS89); at harvest, the berries had 19.1, 19.4, 18.4, and 21.7 °Babo in CA22, PC23, RA23, and CA23, respectively. At harvest, gray mold incidence [disease incidence (DI), i.e., the percentage of bunches with gray mold] and DS (i.e., the percentage of the bunch surface affected by gray

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Table 1. Plant protection products used in experiments

Active ingredient	Active ingredient concentration	Product and producer	Label dose (g ha^{-1})	Application timing
Aureobasidium pullulans DMS 14941-14940 (APD)	2.5 × 10^9 CFU/g per strain	Botector, Manica S.p.a, Rovereto (TN), Italy	400	A
Bacillus amyloliquefaciens D747 (BAD)	5 × 10^{10} CFU/g	Amylo-X, CBC (Europe) S.r.l., Grassobbio (BG), Italy	2000	C
Boscalid	500 g/kg	Cantus, BASF, Cesano Maderno (MB), Italy	1000	C
Cyprodinil + Fludioxonil	375 + 250 g/kg	Switch, Syngenta, Milano (MI), Italy	800	B, D
Fluazinam	500 g/L	Nando Maxi, Sumitomo Chemical Italia S.r.l., Milano (MI), Italy	1000	A
Metschnikowia fructicola NRRL Y-27328 (MFN)	1–3 × 10^{10} cells/g	Noli, Koppert Italia S.r.l., Bussolengo (VR), Italy	2000	D
Trichoderma atroviride SC1 (TAS)	1 × 10^{13} CFU/granule	Vintec, Belchim Crop Protection, Milano (MI), Italy	200	B
Trichoderma gamsii ICC 080 and *Trichoderma asperellum* ICC 012 (TGA)	3 × 10^7 CFU/g	Remedier, Gowan Italia, Faenza (RA), Italy	1000	B

mold), based on the standard diagram of Hill *et al.*,[32] were visually assessed on 100 random bunches per plot in four replicate plots per vineyard.

To assess the presence of latent infection in harvested berries, 25 berries per plot were randomly collected with their pedicel, which did not show any symptoms of rot. Berries were transported to the laboratory in a cooler and were rinsed under tap water for 3 min, disinfested with two-thirds of distilled water and one-third of 5% sodium hypochlorite to remove epiphytic microflora, and finally rinsed again with sterile water. Berries were placed over a metallic grid net mounted in a metallic box (20 cm × 15 cm, with wet filter paper placed at the bottom) so that berries did not touch each other or the filter paper. Each of the four replicate boxes contained 25 berries. The boxes were sealed in plastic bags to maintain a saturated atmosphere and were incubated at 25 °C with a 12 h photoperiod for 10 days. DI and DS were then assessed, and average severity in each box was then calculated based on Hill *et al.*'s recommendations.[32]

2.5 Data analysis

For data analysis, DS observed in the field and after incubation in the laboratory were summed up to account for total disease (i.e., visible and latent). An analysis of variance (ANOVA) was performed to determine whether DS was significantly affected by the epidemics (CA22, CA23, PC23, and RA23), treatments [calendar-based BCA application (BCA calendar), model-based BCA application (BCA model), calendar-based fungicide application (CHEM calendar), model-based fungicide application (CHEM model)], and their interactions. DS data were transformed prior to performing ANOVA by using the inverse sine function to stabilize variances and were then back-transformed using the sine function. Means were compared using the Fisher's protected least significant difference test at $P = 0.05$. Data were analyzed with R software (version 4.3.1; R Core Team, 2023) and R 'agricolae' package.[33]

3 RESULTS

In CA22, the first infection period [i.e., from GS53 (inflorescences clearly visible) to GS73 (berries groat-sized, bunches begin to hang)] lasted 53 days (from 23 April to 14 June). During this period, the average temperature and RH were 19.9 °C (minimum 11.4 °C and maximum 26.8 °C) and 65.6% (minimum 35.0% and maximum 100.0%), respectively, with 137 h of wetness and 159.8 mm of rain distributed over 18 rainy days (Fig. 1). The second infection period [i.e., from GS79 (majority of berries touching each other) to GS89 (berries are ripe for harvest)] lasted 80 days (from 26 June to 13 September). During this period, the average temperature and RH were 25.5 °C (minimum 19.8 °C and maximum 30.4 °C) and 56.7% (minimum 31.0% and maximum 96.5%), respectively, with 176 h of wetness and 319.2 mm of rain distributed over 23 rainy days (Fig. 1). The colored bar in Fig. 1 represents the daily model outputs for the gray mold infection in CA22. The season was characterized by mild infection risk except for two periods that occurred 18–20 August and 3–5 September when severe and intermediate infection risks were predicted, respectively (Fig. 1). The dates of treatments performed in CA22 are reported in Fig. 1. In the treatments conducted per calendar-based strategy, BCAs and CHEM were applied on the following dates: 26 May (A), 17 June (B), 3 August (C), and 19 August (D); in the treatments conducted per model-based strategy, BCA and CHEM were applied on 19 August (GS89) (Fig. 1).

In CA23, the first infection period [i.e., from GS53 (inflorescences clearly visible) to GS73 (berries groat-sized, bunches begin to hang)] lasted 57 days (from 23 April to 18 June). During this period, the average temperature and RH were 18.1 °C (minimum 11.8 °C and maximum 24.5 °C) and 74.2% (minimum 45.9% and maximum 100.0%), respectively, with 311 h of wetness and 160.4 mm of rain distributed over 24 rainy days (Fig. 2). The second infection period [i.e., from GS79 (majority of berries touching each other) to GS89 (berries are ripe for harvest)] lasted 60 days (from 14 July to 11 September). During this period, the average temperature and RH were 18.3 °C (minimum 18.2 °C

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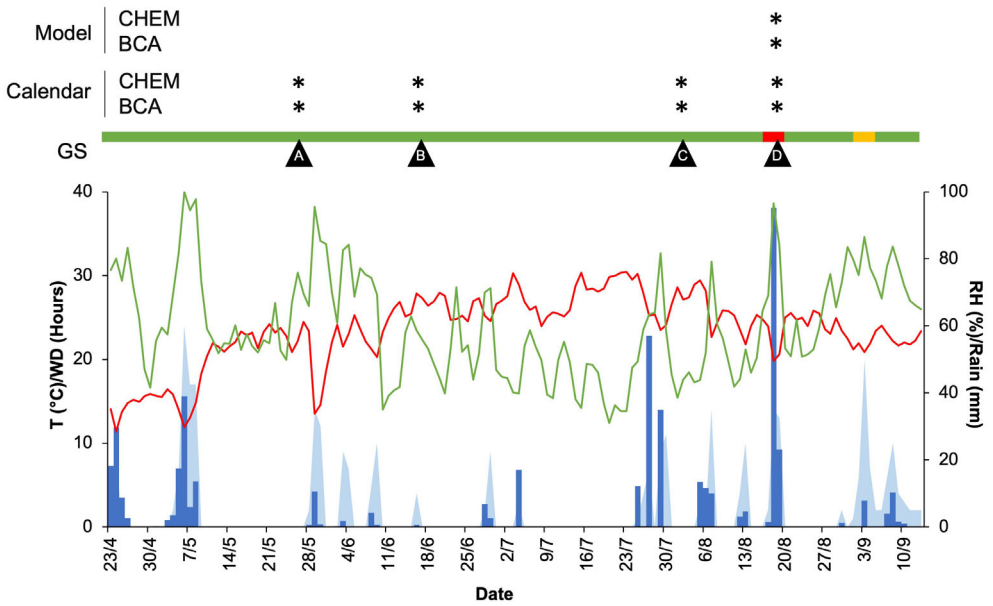


Figure 1. Model output and weather data registered during the epidemic CA22. The horizontal bar represents the classification of gray mold epidemic in CA22 as mild (green), intermediate (yellow), and severe (red) based on the model of González-Domínguez *et al.*,[9] which was operated daily between growth stage GS53 (inflorescences clearly visible)[30] and GS89 (ripe berries). The black triangles indicate grapevine growth stages: A (flowering, GS65), B (pre-bunch closure, GS77), C (veraison, GS83), and D (before harvest, GS89). BCA calendar-based treatments were: APD at stage A, TAS (2022) and TGA (2023) at stage B, BAD at stage C, and MFN at stage D. CHEM calendar-based treatments were: Fluazinam at stage A, Cyprodinil + Fludioxonil at stage B, Boscalid at stage C, and Cyprodinil + Fludioxonil at stage D. MFN was used in BCA model-based treatment and Cyprodinil + Fludioxonil were used in CHEM model-based treatment. Weather conditions in CA22: daily data of temperature (*T*; red line; in degrees Celsius), relative humidity (RH; green line; in percentage), rain (blue bars; in millimeters), and wetness duration (WD; light blue area; in hours). CA22: Castell'Arquato 2022, GS: growth stage, BCA: biocontrol agent, CHEM: chemical fungicide, APD: *Aureobasidium pullulans* DMS 14941-14940, TAS: *Trichoderma atroviride* SC1, TGA: combination of *Trichoderma gamsii* ICC 080 and *Trichoderma asperellum* ICC 012 (2023), BAD: *Bacillus amyloliquefaciens* D747, MFN: *Metschnikowia fructicola* NRRL Y-27328.

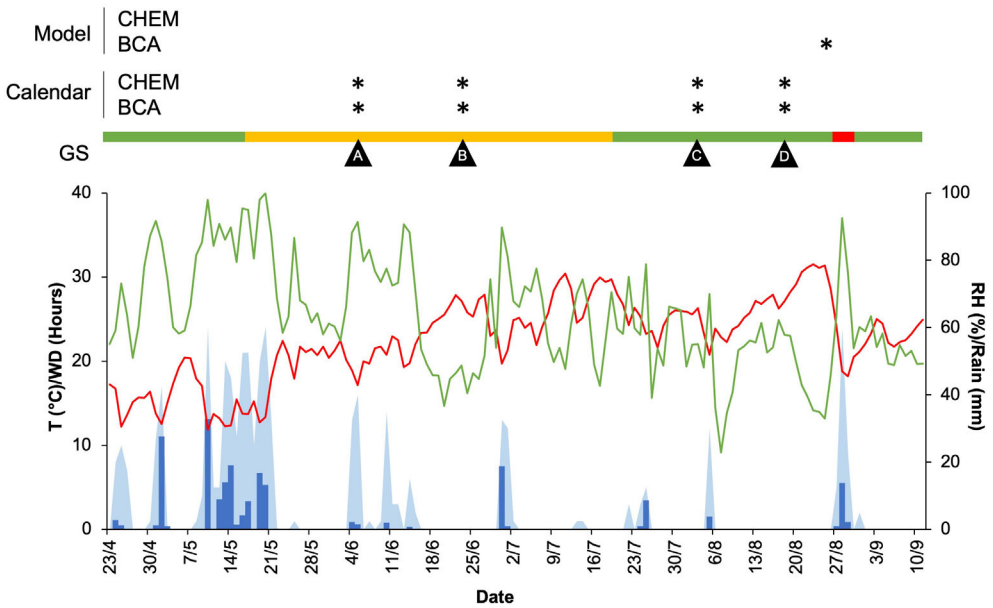


Figure 2. Model output and weather data registered during the epidemic CA23. The horizontal bar represents the classification of gray mold epidemic in CA23 as mild (green), intermediate (yellow), and severe (red) based on the model of González-Domínguez *et al.*,[9] which was operated daily between growth stage GS53 (inflorescences clearly visible)[30] and GS89 (ripe berries). The black triangles indicate grapevine growth stages: A (flowering, GS65), B (pre-bunch closure, GS77), C (veraison, GS83), and D (before harvest, GS89). BCA calendar-based treatments were: APD at stage A, TAS (2022) and TGA (2023) at stage B, BAD at stage C, and MFN at stage D. CHEM calendar-based treatments were: Fluazinam at stage A, Cyprodinil + Fludioxonil at stage B, Boscalid at stage C, and Cyprodinil + Fludioxonil at stage D. MFN was used in BCA model-based treatment. Weather conditions in CA23: daily data of temperature (*T*; red line; in degrees Celsius), relative humidity (RH; green line; in percentage); rain (blue bars; in millimeters), and wetness duration (WD; light blue area; in hours). CA23: Castell'Arquato 2023, GS: growth stage, BCA: biocontrol agent, CHEM: chemical fungicide, APD: *Aureobasidium pullulans* DMS 14941-14940, TAS: *Trichoderma atroviride* SC1, TGA: combination of *Trichoderma gamsii* ICC 080 and *Trichoderma asperellum* ICC 012 (2023), BAD: *Bacillus amyloliquefaciens* D747, MFN: *Metschnikowia fructicola* NRRL Y-27328.

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and maximum 31.5 °C) and 54.7% (minimum 22.9% and maximum 92.5%), respectively, with 64 h of wetness and 30.6 mm of rain distributed over 7 rainy days (Fig. 2). The daily model outputs for the gray mold infection in CA23 are presented in Fig. 2. The season was characterized by mild infection risk until 17 May; a long period of intermediate risk of infection occurred between 18 May and 19 July (Fig. 2). From 20 July to 27 August, the infection risk returned mild; in contrast, a high risk of infection was predicted by the model between 28 and 30 August (Fig. 2). Figure 2 also shows the dates of treatments performed during the season in CA23. In the treatments conducted per calendar-based strategy, BCAs and CHEM were applied on the following dates: 5 June (A), 23 June (B), 3 August (C), and 18 August (D); in the treatments conducted per model-based strategy, BCA was applied on 25 August (> GS89) (Fig. 2). CHEM was not applied following the model-based strategy to respect the preharvest interval of the selected product.

In PC23, the first infection period [i.e., from GS53 (inflorescences clearly visible) to GS73 (berries groat-sized, bunches begin to hang)] lasted 46 days (from 25 April to 9 June). During this period, the average temperature and RH were 18.6 °C (minimum 13.1 °C and maximum 23.0 °C) and 77.2% (minimum 59.1% and maximum 97.4%), respectively, with 200 h of wetness and 144.8 mm of rain distributed over 20 rainy days (Fig. 3). The second infection period [i.e., from GS79 (majority of berries touching each other) to GS89 (berries are ripe for harvest)] lasted 62 days (from 6 July to 5 September). During this period, the average temperature and RH were 26.5 °C (minimum 19.3 °C and maximum 31.4 °C) and 62.8% (minimum 33.0% and maximum 92.3%), respectively, with 49 h of wetness and 62.8 mm of rain distributed over 8 rainy days (Fig. 3). The daily model outputs for the gray mold infections in PC23 are presented in Fig. 3. The season was characterized by mild infection risk until 14 May; a long period of intermediate risk of infection occurred between 15 May and 11 July (Fig. 3). From 12 July to 27 August, the infection risk returned mild; in contrast, a severe risk of infection was predicted by the model between 28 and 30 August (Fig. 3). Figure 3 also shows the dates of treatments during the season in PC23. In the treatments conducted per calendar-based strategy, BCAs and CHEM were applied on the following dates: 26 May (A), 15 June (B), 26 July (C), and 18 August (D); in the treatments conducted per model-based strategy, BCA was applied on 25 August (> GS89) (Fig. 3). CHEM was not applied following the model-based strategy to respect the preharvest interval of the selected product.

In RA23, the first infection period [i.e., from GS53 (inflorescences clearly visible) to GS73 (berries groat-sized, bunches begin to hang)] lasted 47 days (from 5 May to 20 June). During this period, the average temperature and RH were 19.3 °C (minimum 13.9 °C and maximum 26.0 °C) and 79.3% (minimum 55.3% and maximum 95.8%), respectively, with 375 h of wetness and 214 mm of rain distributed over 17 rainy days (Fig. 4). The second infection period [i.e., from GS79 (majority of berries touching each other) to GS89 (berries are ripe for harvest)] lasted 47 days (from 14 July to 9 September). During this period, the average temperature and RH were 24.8 °C (minimum 18.7 °C and maximum 29.4 °C) and 71.1% (minimum 52.4% and maximum 89.4%), respectively, with 264 h of wetness and 38.2 mm of rain distributed over 11 rainy days (Fig. 4). The daily model outputs for the gray mold infection risk in RA23 are presented in Fig. 4. The season was characterized by mild infection risk periods alternated with

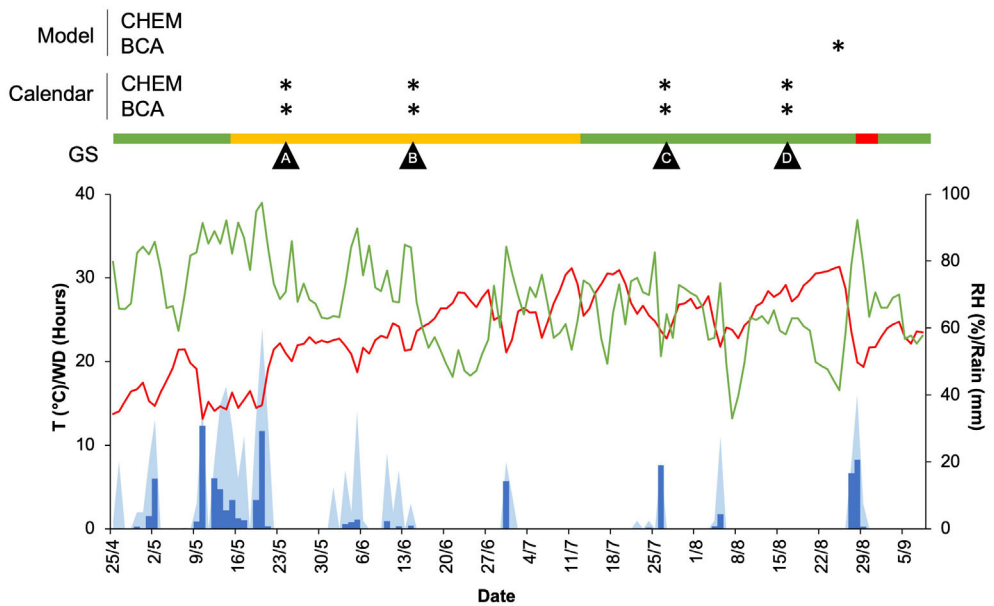


Figure 3. Model output and weather data registered during the epidemic PC23. The horizontal bar represents the classification of gray mold epidemic in PC23 as mild (green), intermediate (yellow), and severe (red) based on the model of González-Domínguez *et al.*,[9] which was operated daily between growth stage GS53 (inflorescences clearly visible)[30] and GS89 (ripe berries). The black triangles indicate grapevine growth stages: A (flowering, GS65), B (pre-bunch closure, GS77), C (veraison, GS83), and D (before harvest, GS89). BCA calendar-based treatments were: APD at stage A, TAS (2022) and TGA (2023) at stage B, BAD at stage C, and MFN at stage D. CHEM calendar-based treatments were: Fluazinam at stage A, Cyprodinil + Fludioxonil at stage B, Boscalid at stage C, and Cyprodinil + Fludioxonil at stage D. MFN was used in BCA model-based treatment. Weather conditions in PC23: daily data of temperature (*T*; red line; in degrees Celsius), relative humidity (RH; green line; in percentage); rain (blue bars; in millimeters), and wetness duration (WD; light blue area; in hours). PC23: Piacenza 2023, GS: growth stage, BCA: biocontrol agent, CHEM: chemical fungicide, APD: *Aureobasidium pullulans* DMS 14941-14940, TAS: *Trichoderma atroviride* SC1, TGA: combination of *Trichoderma gamsii* ICC 080 and *Trichoderma asperellum* ICC 012 (2023), BAD: *Bacillus amyloliquefaciens* D747, MFN: *Metschnikowia fructicola* NRRL Y-27328.

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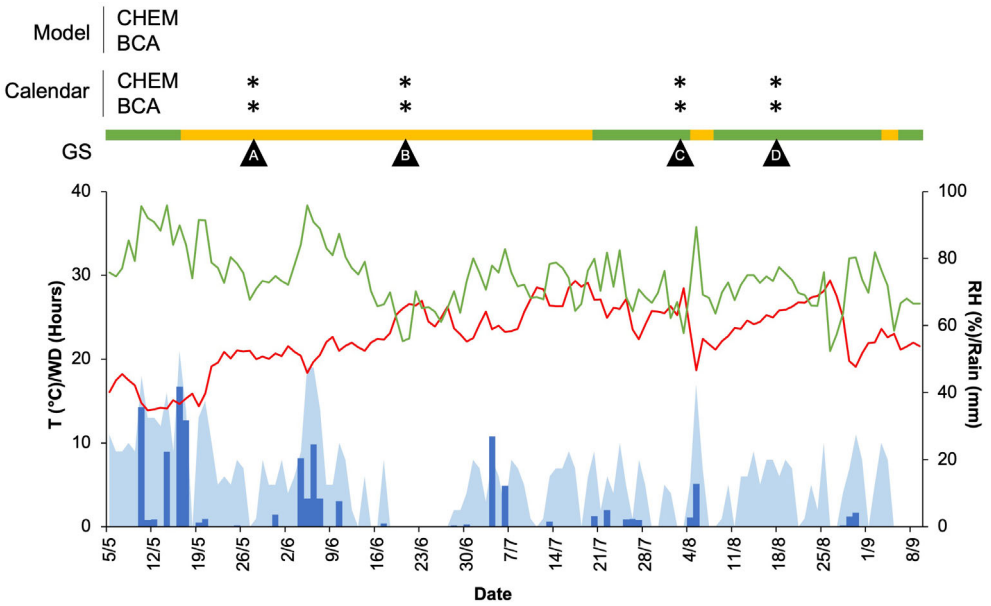


Figure 4. Model output and weather data registered during the epidemic RA23. The horizontal bar represents the classification of gray mold epidemic in RA23 as mild (green), intermediate (yellow), and severe (red) based on the model of González-Domínguez *et al.*,[9] which was operated daily between growth stage GS53 (inflorescences clearly visible)[30] and GS89 (ripe berries). The black triangles indicate grapevine growth stages: A (flowering, GS65), B (pre-bunch closure, GS77), C (veraison, GS83), and D (before harvest, GS89). BCA calendar-based treatments were: APD at stage A, TAS (2022) and TGA (2023) at stage B, BAD at stage C, and MFN at stage D. CHEM calendar-based treatments were: Fluazinam at stage A, Cyprodinil + Fludioxonil at stage B, Boscalid at stage C, and Cyprodinil + Fludioxonil at stage D. No sprays were applied in BCA and CHEM model-based treatments. Weather conditions in RA23: daily data of temperature (*T*; red line; in degrees Celsius), relative humidity (RH; green line; in percentage); rain (blue bars; in millimeters), and wetness duration (WD; light blue area; in hours). RA23: Ravenna 2023, GS: growth stage, BCA: biocontrol agent, CHEM: chemical fungicide, APD: *Aureobasidium pullulans* DMS 14941-14940, TAS: *Trichoderma atroviride* SC1, TGA: combination of *Trichoderma gamsii* ICC 080 and *Trichoderma asperellum* ICC 012 (2023), BAD: *Bacillus amyloliquefaciens* D747, MFN: *Metschnikowia fructicola* NRRL Y-27328.

intermediate infection risk periods until harvest and no severe infection risks were predicted (Fig. 4). Figure 4 also shows the dates of treatments during the season in RA23. In the treatments conducted per calendar-based strategy, BCAs and CHEM were applied on the following dates: 29 May (A), 21 June (B), 3 August (C), and 21 August (D); in the model-based strategy, BCAs and CHEM were not applied during the season (Fig. 4).

The ANOVA showed that epidemics, treatments, and their interactions significantly ($P < 0.001$) affected DS, accounting for 35.7%, 41.8%, and 22.5% of total variance, respectively.

In CA22, NT control showed an overall DS (visible + latent) of 15.7% with bunch areas affected with gray mold (with latent infection accounting for 89.2% of overall DS), which was significantly higher than that observed in any treatment, with no differences in DS after BCA and CHEM treatments following model (one spray) and calendar-based strategies (four sprays) (Fig. 5(A)). In CA23, NT control showed an overall DS (visible + latent) of 3% with bunch areas affected with gray mold, and DS due to latent infection was 2.9%; DS was similar in the CHEM model that was not treated with fungicides because the only infection risk predicted by the model occurred during the preharvest interval (Fig. 2, Fig. 5(B)); DS in the other treatments was significantly lower than in NT control, with the plants treated following the CHEM calendar-based strategy (0.2% DS, with four sprays) being less affected than those treated following the BCA model-based strategy (1.1% DS, with one spray). In PC23, results obtained were similar to those obtained in CA23, with the only difference that the effects of CHEM calendar-based strategy (0.5% DS, with four sprays) and BCA model-based strategy (0.9% DS, with one spray) were not significantly different (Fig. 5(C)). In RA23, overall DS was 4.3% in NT control (largely due to latent infection). As the model did not predict any severe infection all season long, no sprays were applied in CHEM and BCA model-based treatments, which thus showed similar DS. Therefore, the effects of these treatments were not significantly different from that of NT control (Fig. 5(D)). DS in CHEM and BCA calendar-based treatments (four sprays) was 1.9% and 1.3% lower than in NT control, respectively ($P < 0.05$).

Apart from CA22, overall DS was lower than the acceptance threshold of 5%, meaning that disease control was not economically justified in three out of four epidemics. In these three epidemics, the use of the model saved 100% and 83% of fungicide and BCA sprays, respectively. In CA22, the model enabled 75% saving of both fungicide and BCA sprays while lowering DS below the threshold value.

4 DISCUSSION

In the present study, we combined a target-based application of BCAs with disease risk predictions provided by a weather-driven mathematical model. The concept of target-based use of BCAs has its roots in the works of various authors.[15,25,28] It consists of selecting BCAs for a specific intervention based on the plant GS and part to be protected (e.g., flowers, bunch trash, or ripening berries), the *Botrytis cinerea* infection pathway (e.g., conidial infection of flowers or berries, berry-to-berry infection), the BCAs MoA that is required for that infection pathway (e.g., competition, antibiosis, induced resistance). The ecological niche of the BCA is also a part of this concept.[23]

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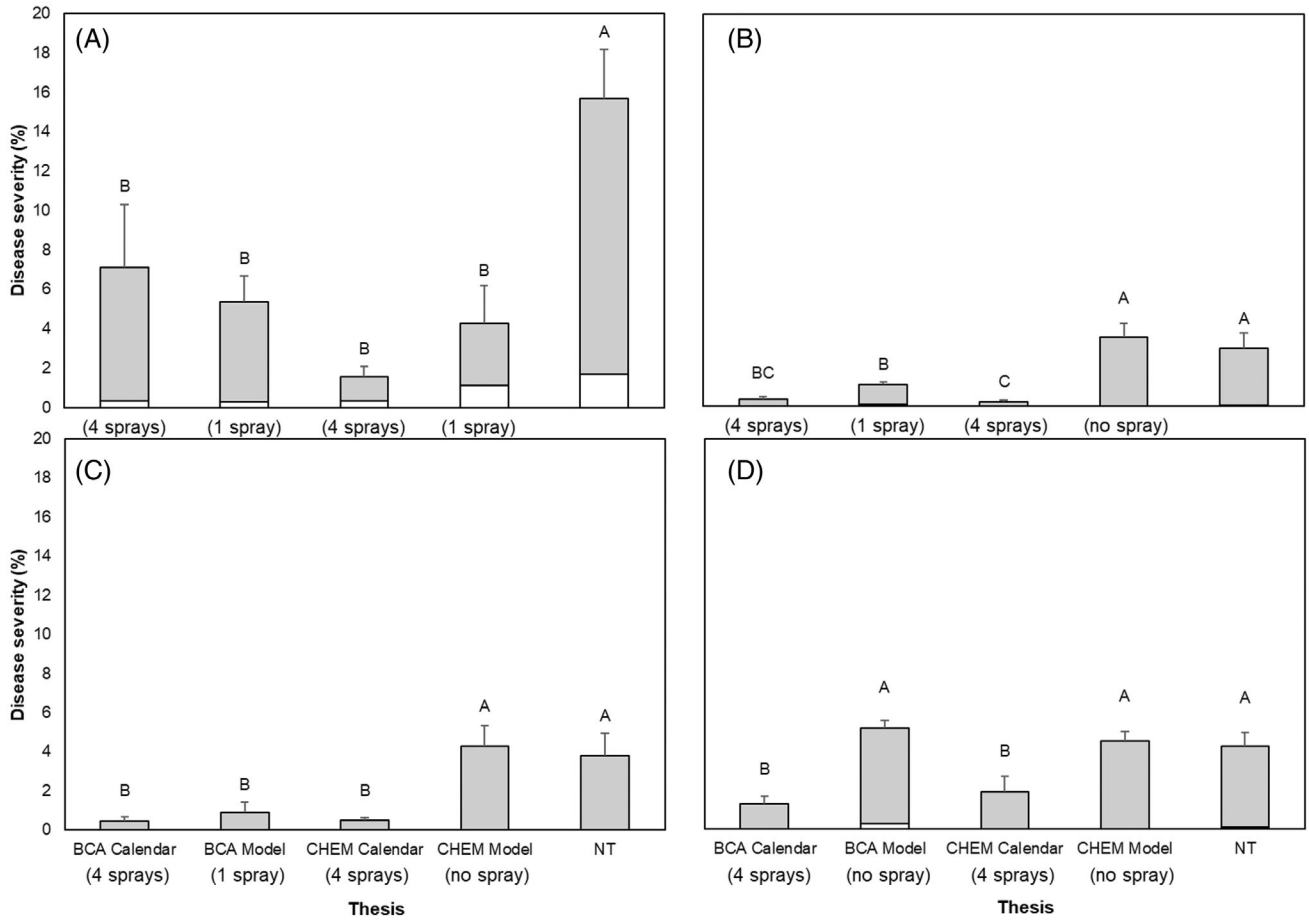


Figure 5. Disease severity assessed at harvest and for latent infections. Bars represent average gray mold disease severity in visible (white bars) and latent infection (light gray bars) at Castell'Arquato 2022 (A), Castell'Arquato 2023 (B), Piacenza 2023 (C), and Ravenna 2023 (D) after following treatments: (i) non-treated (NT) control, (ii) plots treated with chemical fungicides following calendar-based strategy (CHEM calendar), (iii) plots treated with BCAs following calendar-based strategy (BCA calendar), (iv) plots treated with chemical fungicides following model-based strategy (CHEM model), and (v) plots treated with BCAs following model-based strategy (BCA model). In parentheses, the number of sprays applied per treatment is mentioned. Whiskers indicate standard error. Disease severity of latent infection was assessed after 10 days of berries incubation at 25 °C in a saturated atmosphere, disease severity in field was assessed at harvest (> GS89). Significantly different values within each treatment are indicated by different letters according to the least-square difference test at $P < 0.05$.

Following the target-based concept, we used *A. pullulans* at the flowering stage to control conidial infection of flowers. This microorganism competes with *Botrytis cinerea* for nutrients present on plant surfaces, thereby preventing germination of *Botrytis cinerea* and latent infection.[28] To treat infection at pre-bunch closure, *T. atroviride* SC1 was used in 2022 and a combination of *T. gamsii* ICC 080 and *T. asperellum* ICC 012 was used in 2023. All these fungi share the same MoA, that is, colonization of dead plant tissues and competition with *Botrytis cinerea* for space and nutrients.[34] *Bacillus amyloliquefaciens* used to control infection at the veraison stage has antibiosis as the main MoA.[35] To treat preharvest infections, *M. fructicola* was used, which competes for nutrients and space with *Botrytis cinerea*.[36]

The BCAs were applied based on either the plant GSs (calendar-based), specifically at flowering, pre-bunch closure, veraison, and preharvest or the disease risk predicted by a weather-driven model developed by González-Domínguez *et al.*[9] (model-based), and the number of BCA sprays depends on weather conditions. Our results showed that the use of the model helped in achieving a similar level of disease control as the one achieved by using calendar-based strategy, while saving 75% to 100% of unjustified sprays. In previous studies, González-Domínguez *et al.*[2,37] demonstrated that the four-spray calendar strategy can lead to unjustified treatments that are recommended only when severe disease risk is expected; in mild disease risk conditions, a lower number of sprays can provide efficient gray mold control.

Our results were obtained in mild disease conditions; however, this does not reduce the relevance of our study as mild epidemics are more frequent than intermediate and severe epidemics. In two independent datasets of 21 and 23 epidemics that occurred between 1997 and 2018 across Europe, the epidemics with DS < 5% accounted for 42.8% and 69% of the cases, respectively.[9,11] In addition, the frequency of mild epidemics is likely to increase due to climatic changes.[38] Even though the increased frequency of intense rainy periods in spring (often during grapevine flowering, as in our study) would favor the establishment of latent infection in berries, the corresponding increase in dry and hot summer weather (which is the period of grape ripening and harvest)[39] would reduce the onset of latent infection (i.e., the transition from latent infection to visible rot) and disease development due to berry infection by conidia and berry-to-berry infection.[40,41] The mentioned increase in the frequency of latent

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infection is relevant,[42,43] even though it does not induce the appearance of visible symptoms in the field due to climate change, as observed in our study. Indeed, latent infection increases glycerol and gluconic acid content in berries,[44] which is an indicator of poor quality of bunches for winemaking.[45] Therefore, it is expected that vine growers would more frequently face disease management under mild epidemics with latent infections; in such scenarios, the present study could be of help in the decision-making process. The use of the weather-driven model of González-Domínguez *et al.*[9] would represent a useful tool for adapting disease management under climate change;[46] in our study, the model use reduced the number of sprays for the control of *Botrytis cinerea* by 75% to 100% depending on the epidemic. Hence, our study presented an innovative approach to the control of gray mold in vineyards. Further studies are needed to verify whether the same approach is valid under high disease pressure.

In addition, our results confirmed that BCAs are an interesting tool for preharvest treatments.[22] In two epidemics, CA23 and PC23, preharvest sprays in the model-based treatment were applied when using BCAs but not when using CHEM, owing to the longer preharvest interval of the latter. As biochemical and structural changes make berries highly susceptible to gray mold at late ripening stages,[47–49] the failure to apply fungicides close to harvest made DS significantly higher in CHEM-treated berries than in BCA-treated berries.

As the ecological niche of BCAs is also a part of the target-based use of BCAs,[23] further studies are needed to integrate the response of BCAs to weather conditions and forecasts at the time of BCA application in the field, with an objective to apply BCAs that are more suited to temperature and moisture conditions expected, thereby increasing the probability of BCAs to efficiently control pathogens. Finally, the integrated results obtained in the present study can be implemented in a BCA model-based infection control strategy considering the specificity of the BCA, pathogen, host plant, and environment to improve BCA application in fields, counting all variables that can influence BCA efficacy in controlling gray mold in vineyards.[50]

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CONFLICT OF INTEREST STATEMENT

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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